

Hydra

Introduction

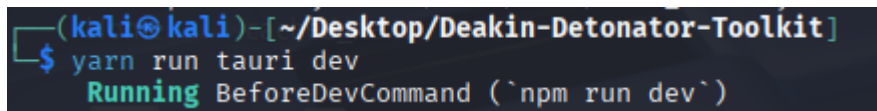
Hydra is an authentication testing tool that targets login mechanisms via a brute-force credential cracking process. Hydra can target myriad protocols including HTTP, FTP, SSH, SMB, and RDP. Hydra supports the implementation of dictionaries that contain a significant volume of passwords.

Disclaimer

Only run this tool against networks, systems and files that you own or those that you have explicit written permission to perform security assessments against. Attempting unauthorised access to third party systems without explicit permission is unethical and illegal.

Setup and launch

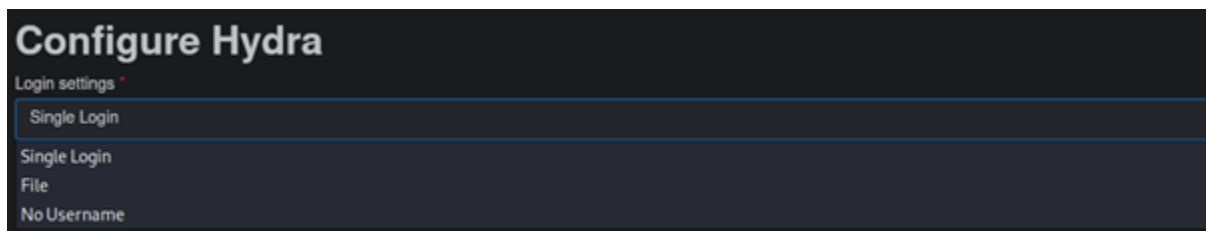
Initialise your virtual machine or device and open the terminal. Navigate to the directory where the Deakin Detonator Toolkit is installed. Then enter the following command into the terminal “yarn run tauri dev”.



```
(kali@kali)-[~/Desktop/Deakin-Detonator-Toolkit]
$ yarn run tauri dev
Running BeforeDevCommand (`npm run dev`)
```

Once the startup process has completed accept the user agreement and then navigate to the upper left hand of the Deakin Detonator Toolkit GUI and click on the tools button. Once in the tool selection screen select Hydra or use the quick search feature to access Hydra, then proceed to the tool's configuration panel.

Configuration and Deployment

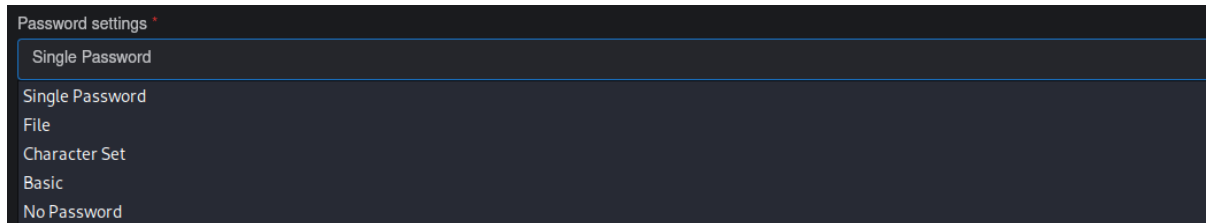


Once Hydra has initialised the first option available is which login mechanism to target. The single login option applies to a username and is either locally accessible or a remote login mechanism which. Should the Single login option be selected then the username must be entered into the following field:



The file option allows a user to upload a dictionary of usernames to attempt in the credential cracking process. When using DDT, the file must be stored in the directory Home directory.

The final option is deployed against systems that require no username to access and will cycle through however many passwords are chosen by the user.



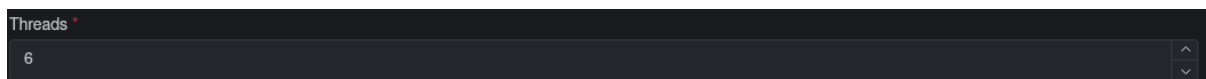
The next step requires selecting which password option best suits the user's testing requirements. Selecting a single password is the simplest and least effective method that is only suitable if the user has a small pool of passwords to work through.

The file option loads a dictionary of potential passwords that are testing one after the other in a brute force manner.

The character set option allows the user to set a custom syntax of passwords to attempt. The user can specify the minimum and maximum word length, whether to use lowercase, uppercase, numbers and/or special characters.

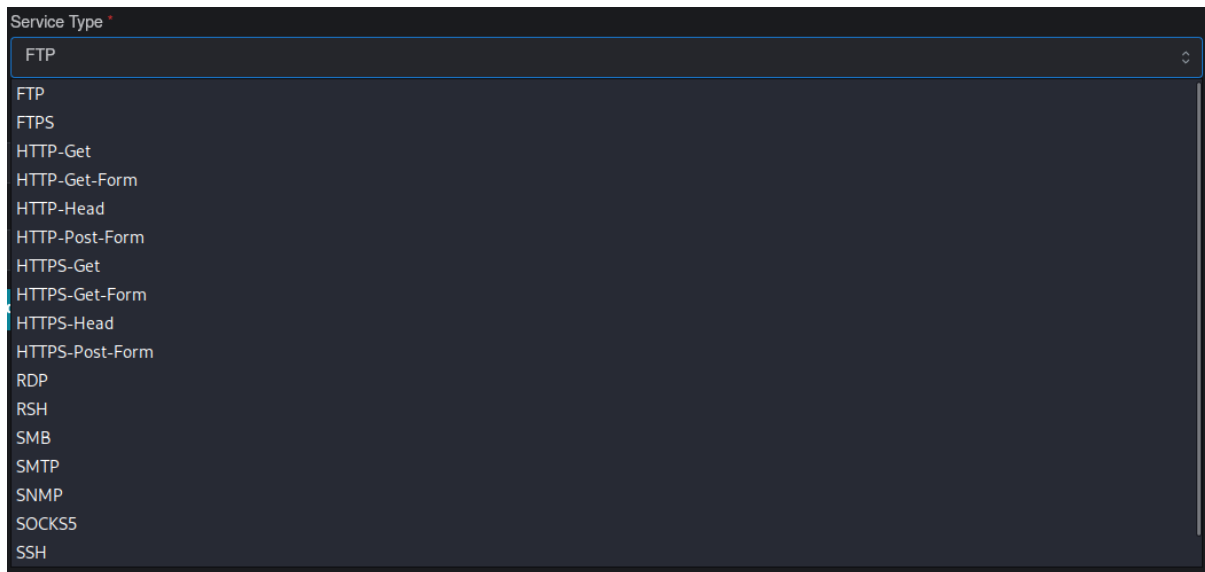
The basic option uses the flags "-e nsr" which will attempt to login while the password is null, will attempt to use the login as the password and attempt a reverse login.

The final option uses no password similarly to the no login option in the first configuration step and targets systems that require only a username to access.



This configuration step requires the user to select the number of CPU threads allocated to Hydra during the cracking process. Before selecting this option determine the number of threads your CPU has and never exceed that number. It is preferable to leave at least 30% of your total thread for OS operations. Allocating more threads to Hydra will reduce the time required to complete the cracking process.

The next step requires the user to select which protocol to use when performing the credential cracking process. This will vary based on the protocol in use by the target. Common protocols in use are FTP, HTTP, HTTPS and SSH.



Service Type *

FTP

FTP

FTPS

HTTP-Get

HTTP-Get-Form

HTTP-Head

HTTP-Post-Form

HTTPS-Get

HTTPS-Get-Form

HTTPS-Head

HTTPS-Post-Form

RDP

RSH

SMB

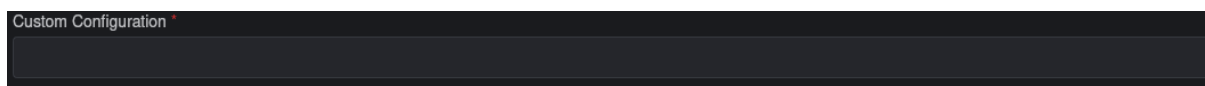
SMTP

SNMP

SOCKS5

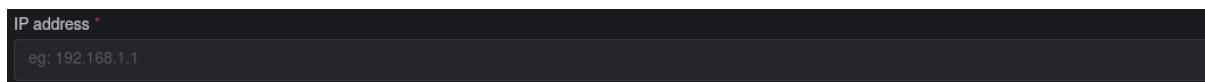
SSH

When selecting any of the HTTP or HTTPS options a new field will appear. This field accepts the specific URL path where the authentication occurs such as “/secure/login” which will be appended to the target’s URL or hostname.



Custom Configuration *

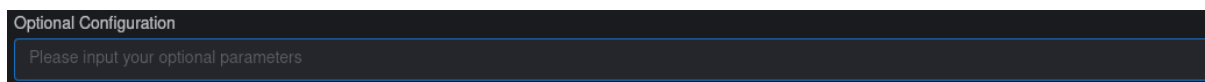
The final mandatory configuration step for the DDT implementation of Hydra requires an IP address. When using Hydra externally an IP address, URL or hostname are all valid targets. Creating a target list file will allow Hydra to brute force multiple targets with the provided credentials and passwords.



IP address *

eg: 192.168.1.1

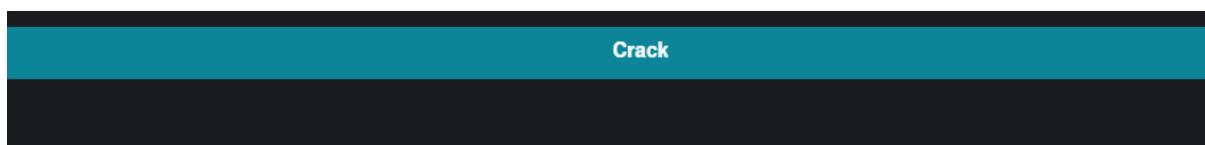
The final field is optional and enables the addition of further flags and parameters to be appended to the command configured in the previous steps.



Optional Configuration

Please input your optional parameters

To complete the configuration process and initiate the credential cracking process click the “Crack” button at the bottom of the configuration page.



Crack

Additional Flags

The custom configuration input can accept multiple flags in addition to the flags set using the GUI configuration options. These include but are not limited to the following flags [1]:

-s	Implements a custom port number entered after “-s”.
-v	Enables verbose output.
-V	Displays username and password for every cracking attempt.
-4	Forces the use of IPv4 addresses.
-6	Forces the use of IPv6 addresses.
-K	When credentials fail to login. Hydra will not use them again.
-c	Limits the time between each login attempt across all threads.
-u	Hydra will try each username with a single password and then move to the next password and loop through all usernames again.
-f	Terminates the process when the first valid set of credentials is discovered.

Output

The output below is expected when set of credentials is not successfully discovered. In this case I used a single username and password, and they failed. Which is to be expected.

```

Output

[DATA] attacking smtp://127.0.0.1:25/

[ERROR] all children were disabled due too many connection errors

0 of 1 target completed, 0 valid password found

Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-09-21 03:31:13

Process terminated with exit code: 255 and signal code: null

```

The output contains vital information even in when the cracking process failed. No passwords were found, and the Error suggests that port 25 (SMTP) is either not open or could have a limit on successive connection attempts.

In the case of a successful cracking attempt the output will be nearly identical the username and password will be displayed. In the case that multiple sets of credentials are successfully discovered then they will also be displayed. Below is a successful output that has been saved to a file using the “Save as Text File” button.

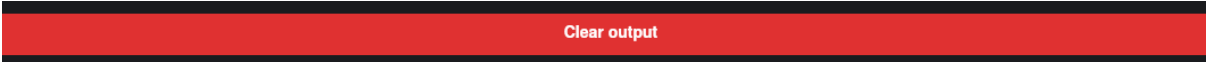
```

PT-020-02-700194721 (Baseline-02-Setup) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
*~Hydra_Test_File.txt - Mousepad
File Edit Search View Document Help
1 $ hydra -l /home/kali/HYDRA -p 1234 -t 4 ftp://192.168.0.1
2
3 Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).
4
5 [INFO] several providers have implemented cracking protection, check with a small wordlist first - and stay legal!
6
7 Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-09-21 03:49:15
8
9 [DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), -1 try per task
10
11 [DATA] attacking ftp://192.168.0.1:21/
12
13 [VERBOSE] 192.168.0.1:21|login 'hello' - pass '1234'
14
15 [80][ftp] host: 192.168.0.1 login: hello password: 1234
16 1 of 1 target successfully completed, 1 valid password found
17
18 Process completed successfully.

```

Save as Text File (.txt)

To clear the output and start the process again, click the clear output button and then select a new target, ensuring to clear the current fields manually or by closing Hydra and selecting the tool once again from the tool selection screen.



Troubleshooting

The most common issue encountered with Hydra is poor-quality dictionary files. Ensure the dictionary files are properly formatted. This is especially critical when using multiple dictionary files as it may be difficult to determine exactly which one is misconfigured. It is best to test examine each file for errors individually and then proceeding with the penetration test.

Port configuration is the next most common issue. Using default port numbers is unlikely to succeed, many organisations and network administrators configure services to use alternate port numbers. It is critical to perform appropriate reconnaissance using a tool such as NMAP to determine what services are running on the targeted host and what port numbers those services are associated with before attempting a penetration test with Hydra.

Be aware that many systems are configured to limit or halt login attempts after a certain volume of attempts. This will greatly limit the chance of success and potentially alert the target to the credential cracking attempt.

References

- [1] Offensive Security, "Tool Documentation: Hydra," Offensive Security, 2025. [Online]. Available: <https://www.kali.org/tools/hydra/>. [Accessed 11 September 2025].